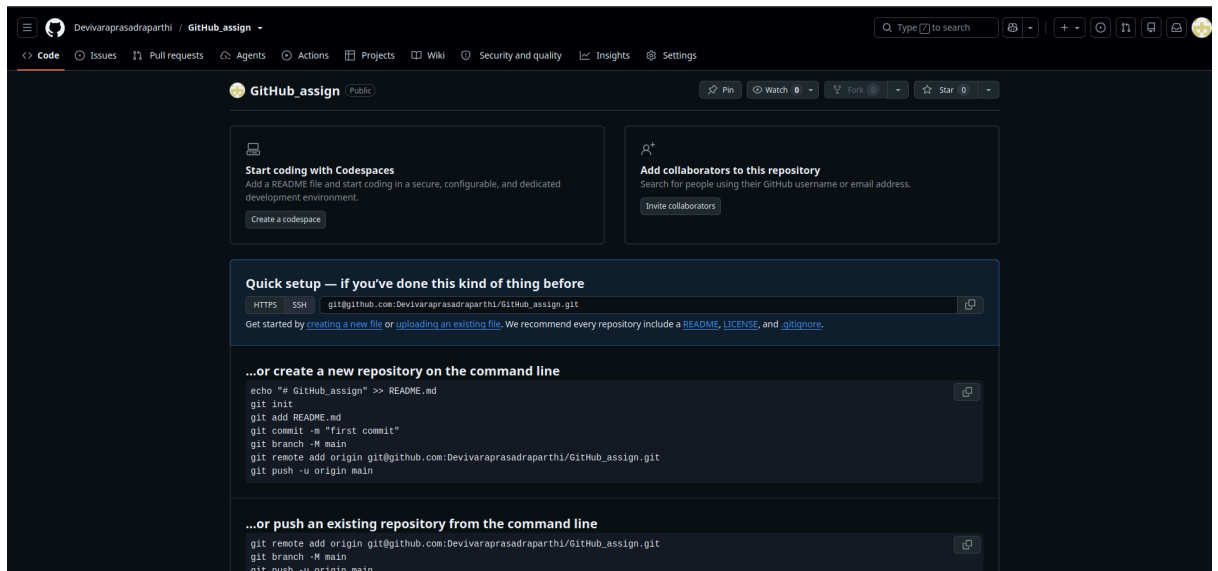


Git & GitHub

Assignment-4

Task 1: Repository Setup & First Branch

1. Create a new GitHub repository.



2. Clone the repository to your local machine using SSH (generate an SSH key if needed, and add the public key to your GitHub account).

```
● dvprasad@ubuntu24:~/Desktop/Git&GitHub_dvp$ git clone git@github.com:Devivaraprasadraparthi/GitHub_assign.git
Cloning into 'GitHub_assign'...
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvC0qU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Warning: You appear to have cloned an empty repository.
● dvprasad@ubuntu24:~/Desktop/Git&GitHub_dvp$ git status
On branch master

No commits yet

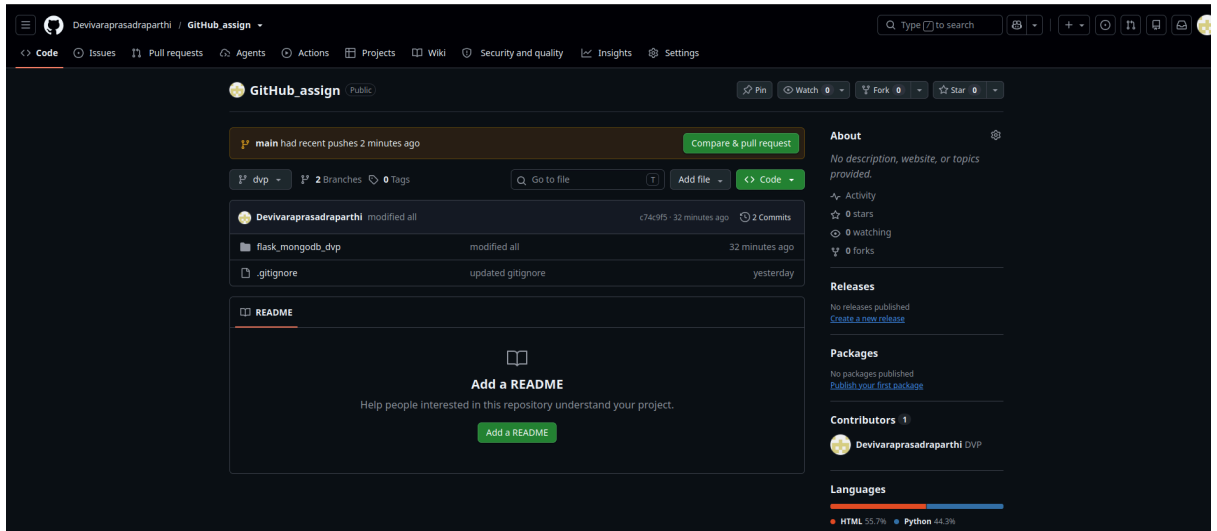
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    GitHub_assign/

nothing added to commit but untracked files present (use "git add" to track)
○ dvprasad@ubuntu24:~/Desktop/Git&GitHub_dvp$
```

3. Create a new branch named after your username (e.g., Tutedude).

```
do not check if another worktree is holding the given r
• dvprasad@ubuntu24:~/Desktop/Git&GitHub_dvp/GitHub_assign$ git switch -c dvp
Switched to a new branch 'dvp'
• dvprasad@ubuntu24:~/Desktop/Git&GitHub_dvp/GitHub_assign$ git branch
```

4. Add your Flask project files to this branch.



5. Commit the changes and merge the branch into the main branch.

```
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ ls -la
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git push -t origin main
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 6 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (13/13), 1.79 KiB | 918.00 KiB/s, done.
Total 13 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 2 local objects.
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/Devivaraprasadraparthi/GitHub_assign/pull/new/main
remote:
To github.com:Devivaraprasadraparthi/GitHub_assign.git
 * [new branch]      main -> main
💡 dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

Task 2: Update JSON & Resolve Conflicts

6. Create a new branch named `_new` (e.g., `Tutedude_new`).

```
Problems 4 Output Debug Console Terminal Ports
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git branch
  dvp
  * main
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git checkout -b dvp_new
Switched to a new branch 'dvp_new'
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git branch
  dvp
  * dvp_new
  main
○ dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

7. Update the content of the JSON file used for the /api route in this branch.

```
Problems 4 Output Debug Console Terminal Ports
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
(use "git restore <file>..." to discard changes in working directory)
  modified:   flask_mongodb_dvp/app.py

no changes added to commit (use "git add" and/or "git commit -a")
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git add .
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
On branch dvp_new
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   flask_mongodb_dvp/app.py
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git commit -m "updated JSON"
[dvp_new daf7b7b] updated JSON data
1 file changed, 1 insertion(+)
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
```

8. Merge the _new branch into the main branch.

```
Problems 4 Output Debug Console Terminal Ports
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git checkout main
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git checkout main
Switched to branch 'main'
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
On branch main
nothing to commit, working tree clean
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git branch
  dvp
  dvp_new
  * main
• dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git merge dvp_new
Updating flc0d23..daf7b7b
Fast-forward
 flask_mongodb_dvp/app.py | 1 +
 1 file changed, 1 insertion(+)
○ dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

9. If there are conflicts during the merge, resolve them by accepting the changes from the _new branch.
10. Add the resolved changes to the staging area, commit them, and push the updates to the remote repository.

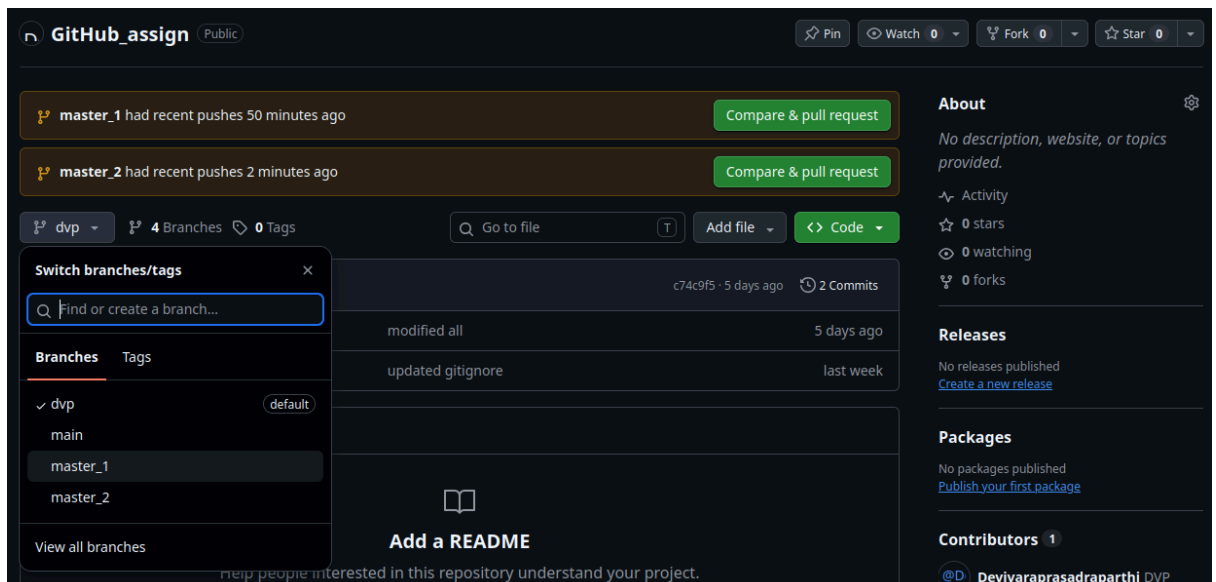
```
Problems 4 Output Debug Console Terminal Ports

dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git push
To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

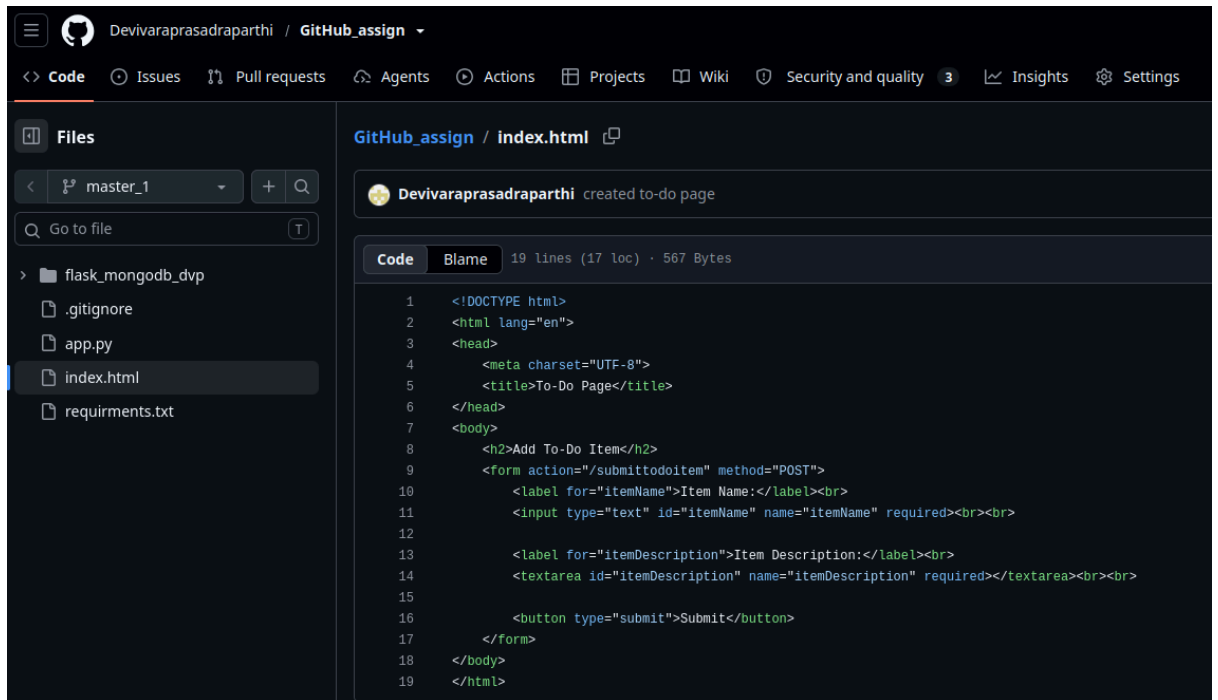
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git push --set-upstream ori
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 6 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 505 bytes | 505.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Devivaraprasadraparthi/GitHub_assign.git
    f1c0d23..daf7b7b  main -> main
branch 'main' set up to track 'origin/main'.
○ dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

Task 3: Parallel Feature Development

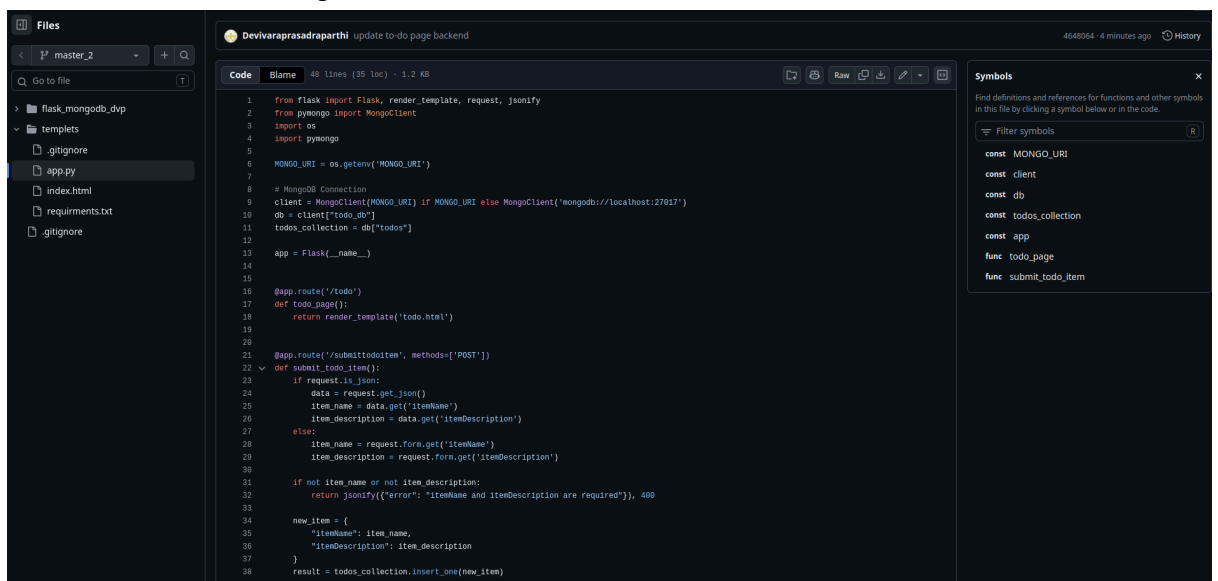
11. **Branch Creation:** Create two branches, master_1 and master_2, from the main branch.



12. **Feature Development in master_1:** Create a To-Do Page in the frontend with a form containing the fields Item Name and Item Description.



13. **Backend API in master_2:** Create a backend route named /submittodoitem that accepts itemName and itemDescription via a POST request and stores these details in a MongoDB database.



14. **Merging Changes:** Merge the changes from both master_1 and master_2 into the main branch.

```

● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git merge master_1 -m "Merge master_1 into main"
Updating daf7b7b..445524a
Fast-forward (no commit created; -m option ignored)
 app.py | 5 +++++
 index.html | 19 +++++++++++++++++++++
 requirments.txt | 4 +++
 3 files changed, 28 insertions(+)
 create mode 100644 app.py
 create mode 100644 index.html
 create mode 100644 requirments.txt
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git merge master_1
Already up to date.
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git merge master_2
Updating 445524a..95743b7
Fast-forward
 app.py | 5 -----
 templets/.gitignore | 4 +++
 templets/app.py | 50 +++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 index.html => templets/index.html | 0
 requirments.txt => templets/requirments.txt | 0
 5 files changed, 54 insertions(+), 5 deletions(-)
 delete mode 100644 app.py
 create mode 100644 templets/.gitignore
 create mode 100644 templets/app.py
 rename index.html => templets/index.html (100%)
 rename requirments.txt => templets/requirments.txt (100%)
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
● dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git push
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:Devivaraprasadrparthi/GitHub_assign.git
 daf7b7b..95743b7 main -> main
○ dvprasad@ubuntu24:~/Desktop/GitHub_assign$ 

```

Task 4: Sequential Commits, Reset & Rebase

15. Enhancing the To-Do Form in master_1: Add the following fields to the To-Do form: Item ID, Item UUID, Item Hash.

```

5 | <title>To-Do Page</title>
6 | </head>
7 | <body>
8 |   <h2>Add To-Do Item</h2>
9 |   <form action="/submittodoitem" method="POST">
10 |     <label for="itemName">Item Name:</label><br>
11 |     <input type="text" id="itemName" name="itemName" required><br><br>
12 |
13 |     <label for="itemId">Item ID:</label><br>
14 |     <input type="text" id="itemId" name="itemId"><br><br>
15 |
16 |     <label for="itemUuid">Item UUID:</label><br>
17 |     <input type="text" id="itemUuid" name="itemUuid"><br><br>
18 |
19 |     <label for="itemHash">Item Hash:</label><br>
20 |     <input type="text" id="itemHash" name="itemHash"><br><br>
21 |
22 |     <label for="itemDescription">Item Description:</label><br>
23 |     <textarea id="itemDescription" name="itemDescription" required></textarea><br><br>
24 |
25 |     <button type="submit">Submit</button>
26 |   </form>
27 | </body>
28 | </html>

```

16. **Committing in Sequence:** Add and commit each field separately — first commit: Item ID; second commit: Item UUID; third commit: Item Hash.

```
2 <html lang="en">
7 <body>
9 <form action="/submittodoitem" method="POST">
10
19 <label for="itemHash">Item Hash:</label><br>
20 <input type="text" id="itemHash" name="itemHash"><br><br>
21
22 <label for="itemDescription">Item Description:</label><br>
23 <textarea id="itemDescription" name="itemDescription" required></textarea><br><br>
24
25 <button type="submit">Submit</button>
26 </form>
27 </body>
28 </html>
```

```
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git checkout master_1
Switched to branch 'master_1'
Your branch is up to date with 'origin/master_1'.
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git add .
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
On branch master_1
Your branch is up to date with 'origin/master_1'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   index.html

dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git commit -m "added ITEM ID"
[master_1 6fe22ac] added ITEM ID
1 file changed, 3 insertions(+)
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git add .
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git commit -m "added added item hash"
[master_1 eafb663] added added item hash
1 file changed, 3 insertions(+)
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git add .
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git commit -m "added added UUID item"
[master_1 62a44d4] added added UUID item
1 file changed, 3 insertions(+)
dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

17. **Merging to main:** Merge the master_1 branch into the main branch.

```
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git commit -m "added added item hash"
1 file changed, 3 insertions(+)
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git add .
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git commit -m "added added UUID item"
[master_1 62a44d4] added added UUID item
1 file changed, 3 insertions(+)
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git merge master_1 -m "added itms hash"
Merge made by the 'ort' strategy.
 templates/index.html | 9 ++++++++
1 file changed, 9 insertions(+)
dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

18. **Git Reset:** In the main branch, roll back to the commit where only the Item ID field was added. Use `git reset --soft` to ensure changes remain staged, then re-commit this state and merge it to main.

```
Problems Output Debug Console Terminal Ports
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git status
Your branch and 'origin/main' have diverged,
and have 1 and 3 different commits each, respectively.
(use "git pull" if you want to integrate the remote branch with yours)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    deleted:    app.py
    new file:   templets/.gitignore
    new file:   templets/app.py
    renamed:    index.html -> templets/index.html
    renamed:    requirments.txt -> templets/requirments.txt

dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

19. Rebasing: Rebase the updated main branch changes onto master_1 using git rebase main master_1. Preserve individual commits (do NOT squash) to maintain commit history.

```
valuable there.
dvprasad@ubuntu24:~/Desktop/GitHub_assign$ git rebase --continue
dropping 62a44d4b8cca0f1264095867c12fd1d7411ec023 added added UUID item -- patch contents already upstream
Successfully rebased and updated refs/heads/master_1.
dvprasad@ubuntu24:~/Desktop/GitHub_assign$
```

GitHub REPO LINK:

https://github.com/Devivaraprasadraparthi/GitHub_assign